Amazon Prime Video Catalog Analysis

Content Strategy Case Study

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1. Executive Summary

Streaming platforms compete on the quality and depth of their catalogs. This case study analyzes 9,868 Amazon Prime Video titles and their cast and crew credits to understand how the catalog is composed, how it is rated, what drives popularity and where the strategic gaps are.

The analysis finds a movie-heavy catalog (86.25% movies) with a long middle of average-rated content, a small premium tier (1.68% of titles rated 8 or above on both IMDb and TMDB), a strong skew toward US productions (about 54%) and Drama titles (about 48%), and a clear rating advantage for series over movies (IMDb 7.1 vs 5.8). Four interactive Tableau dashboards present these findings for business users.

2. Business Context and Objective

Context. A streaming catalog is a portfolio of assets. Decisions on content acquisition, licensing renewals, original production and homepage curation all depend on understanding what the catalog contains and how each segment performs.

Objective. Build a decision-support view of the Prime Video catalog that answers:

1. What is the content mix by type, genre, country and age certification?
2. How is the catalog rated, and which segments perform best?
3. Which titles and talent stand out?
4. Where are the gaps and opportunities for content strategy?

3. Data and Tools

Item	Detail
Titles dataset	9,868 titles with type, release year, runtime, genres, production countries, age certification, IMDb and TMDB scores, votes and popularity
Credits dataset	Cast and crew records (actors and directors) linked to titles by title ID
Preparation	Python (pandas) for cleaning and for splitting multi-value genre and country fields
Visualization	Tableau, four dashboards with button-style filters and filter actions

4. Methodology

1. **Data preparation.** Genre and country fields stored as lists were split into one row per title and value, producing separate genre and country tables (22,060 genre rows across 19 genres; 10,248 country rows across 117 countries).
2. **Missing values.** Missing IMDb and TMDB scores were excluded from rating charts instead of being imputed, so that averages reflect real ratings only.

3 **Modeling.** Titles, credits, genres and countries were connected with relationships on title ID, and all title counts use distinct counts to avoid double counting.

4 **Derived fields.** Score bins, runtime bins, a Decade field, Premium % (IMDb and TMDB both 8 or above) and High Rated % (IMDb 8 or above).

5 **Dashboard design.** A consistent dark theme, one color per content type (movie and show), a filter panel (type, age certification, decade) and filter actions linking every chart.

5. KPI Snapshot

KPI	Value
Total titles	9,868
Movie share	86.25% (shows 13.75%)
Premium titles (IMDb and TMDB both 8 or above)	1.68%
High rated titles (IMDb 8 or above)	5.47%
Average IMDb score	Movies 5.8, Shows 7.1
Average TMDB score	Movies 5.8, Shows 7.3

6. Key Findings

1. The catalog is movie-led, but series rate higher. Movies make up 86.25% of titles, yet shows average 7.1 on IMDb against 5.8 for movies. Shows are only 13.75% of the catalog, so this is a signal to investigate, not proof that series always outperform.

2. Quality is concentrated in a small tier. IMDb scores peak near 6.0 and most titles sit between 5 and 7. Only 5.47% of titles score 8 or above on IMDb, and 1.68% reach that level on both IMDb and TMDB.

3. Release volume peaked around 2019 to 2021. Annual releases rose sharply in the late 2010s, with 2021 the highest year (856 titles), followed by 2019 (807) and 2020 (698). Figures for the latest years may be incomplete.

4. Genre mix is concentrated. Drama appears in 4,762 titles (about 48% of the catalog), followed by Comedy (2,987), Thriller (2,119), Action (1,820) and Romance (1,751). Titles can carry several genres, so genre counts add up to more than the total.

5. The catalog is geographically concentrated. The US is a production country for 5,331 titles (about 54%), followed by India (1,072), the UK (928), Canada (530) and France (250).

6. Talent depth reflects classic content. The most credited actors, such as George "Gabby" Hayes, Roy Rogers and Gene Autry, appear in roughly 40 to 50 titles each and are associated with classic Westerns, which points to a large archive of older content.

7. Popularity and quality are not the same thing. The most popular titles by TMDB popularity include All the Old Knives, Suits and Better Call Saul, and they are not simply the highest rated, so curation should consider both measures.

8. Runtime clusters near 90 minutes. The runtime distribution peaks at about 90 minutes, consistent with standard feature length.

7. Strategic Recommendations

#	Recommendation	Evidence	Confidence
1	Evaluate expanding high-quality series	Shows average IMDb 7.1 vs 5.8 for movies, but are only 13.75% of titles	Medium, validate with viewership data
2	Curate and promote a premium tier	Only 1.68% of titles score 8 or above on both IMDb and TMDB	High
3	Review the long middle of mid-rated content at licensing renewal	Most titles score between 5 and 7	Medium
4	Diversify genre and regional mix	Drama is about 48% of titles and the US about 54%	Medium
5	Improve discovery with genre, region, certification and decade filters	Catalog is large and uneven across segments	High

8. Limitations and Risks

- More than 1,000 titles have no IMDb or TMDB score and are excluded from rating analysis.
- Release data for the most recent years appears incomplete, which can make late-period trends look like a decline.
- The data describes the catalog only. It has no viewership, watch time or retention data, so conclusions about engagement or churn cannot be drawn.
- IMDb and TMDB ratings are public opinion proxies and may not reflect Prime Video's own audience.
- Multi-genre and multi-country titles are counted once per value, so segment totals exceed the title count.

9. Next Steps

1 Combine the catalog with viewership and engagement data to test whether higher-rated series also drive more viewing.

2 Benchmark against competitor catalogs (for example Netflix and Disney+) to identify gaps.

3 Track the catalog over time to monitor additions and removals.

4 Add cost and licensing data to estimate the value of each segment.

Appendix: Dashboards

Dashboard	Business question	Key visuals
Company Overview	What does the catalog look like at a glance?	KPI cards, movie vs show split, release trend, IMDb distribution, top actors
Content and Ratings	How is the catalog rated and how long is it?	TMDB and runtime distributions, ratings by type, popular titles
Relationships and Talent	What relationships exist between year, runtime and ratings?	Scatter and bubble plots, IMDb vs TMDB, top actors and directors

Dashboard	Business question	Key visuals
Genres and Countries	Where does the catalog concentrate?	Top genres, genre trends, genres by type, top countries